

Robust Treadable Surface Segmentation of a Simulated 4-Step Staircase Point Cloud Using RANSAC, PCA-Normal Analysis, and DBSCAN

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Abstract—Walking upstairs with legged and tracked robots requires the ability to reliably perceive where a foot could be safely placed. We cast this as a point-cloud segmentation problem where each point of a staircase must be tagged as a *steppable* tread, a vertical *riser*, or *other* (edges, noise, outliers). We create a parametric generator which analytically synthesizes a 4-step staircase, with forced Gaussian surface jitter, isotropic LiDAR noise, and uniform outliers, producing a fully labeled cloud of $\sim 18k$ points. We implement and analyze four conventional, non-learning geometric algorithms on this data: (i) per-point PCA normal analysis, (ii) multi-plane RANSAC, (iii) normal-augmented DBSCAN with per-cluster slope classification, and (iv) height-histogram tread identification. We report Accuracy, Precision, Recall, F1 and IoU for the steppable class, and assess robustness to sensor noise and outliers. The PCA-normal technique yields greatest overall agreement (F1 = 0.913, IoU = 0.840), whereas DBSCAN and height-histogram provide the highest steppable precision (≈ 0.98), which is the metric most directly controlling foothold safety. We propose that a precision-oriented operating point is preferable for a stair-following robot and examine the consequences of the observed noise sensitivity.

Index Terms—point cloud, segmentation, RANSAC, DBSCAN, surface normals, LiDAR, stair-climbing robot, geometric perception

I. INTRODUCTION

WALKING up stairs is a classic problem for mobile robots functioning in human contexts. Before any motion planning, the robot has to transform a raw range scan into a semantic understanding of the geometry, especially *which surfaces can bear a foot* [1]. A tread misclassified as a riser just wastes a candidate foothold, whereas a riser or an edge misclassified as a tread can cause a misstep. Steppable-surface segmentation is asymmetric and safety-critical, i.e., false positives are considerably more expensive than false negatives.

This work gives a systematic, reproducible examination of traditional geometric segmentation for this problem. Point-cloud segmentation has been widely studied [2], and learning-based segmenters such as PointNet [3] now dominate large-scale benchmarks. However, they need labelled training data and offer limited interpretability. Instead of learning from data, we leverage the strong geometric priors of a staircase (big

planar horizontal treads, large vertical risers, thin clutter at their connections) and compare four algorithms that encapsulate these priors in different ways. Our work contributes:

- a parametric generator that generates a 4-step staircase point cloud from explicit equations, with realistic LiDAR noise, outliers and exact ground-truth labels;
- an implementation and head-to-head comparison of four geometric methods (PCA normals, multi-plane RANSAC, normal-augmented DBSCAN, height-histogram);
- a noise- and outlier-sensitivity study of the steppable-class metrics; and
- a precision-oriented safety analysis for foothold selection by a stair-following robot.

II. STAIRCASE POINT-CLOUD GENERATION

We construct a staircase with $N_s = 4$ steps of width w (x-axis), tread depth d (y-axis) and step height h (z-axis). For each step $i \in \{0, \dots, N_s - 1\}$ the two dominant surfaces are sampled as follows.

Tread (steppable, horizontal):

$$\begin{cases} x \sim \mathcal{U}(0, w) \\ y \sim \mathcal{U}(i d, (i+1) d) \\ z = i h + \epsilon_z, \quad \epsilon_z \sim \mathcal{N}(0, \sigma_z^2) \end{cases} \quad (1)$$

Riser (vertical):

$$\begin{cases} x \sim \mathcal{U}(0, w) \\ y = i d + \epsilon_y, \quad \epsilon_y \sim \mathcal{N}(0, \sigma_y^2) \\ z \sim \mathcal{U}((i-1) h, i h) \end{cases} \quad (2)$$

We utilize $w = 2.5$ m, $d = 0.30$ m, $h = 0.18$ m, and $\sigma_z = \sigma_y = 0.02$ m, all within the recommended ranges. On top of the surface samples we add three perturbations representative of real range data:

- 1) *LiDAR noise*: isotropic Gaussian displacement $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma_L^2 I_3)$ applied to every point, with default $\sigma_L = 0.006$ m;
- 2) *Outliers*: a fraction ρ of points drawn uniformly from the staircase bounding volume (default $\rho = 0.03$);

Ground-truth labels

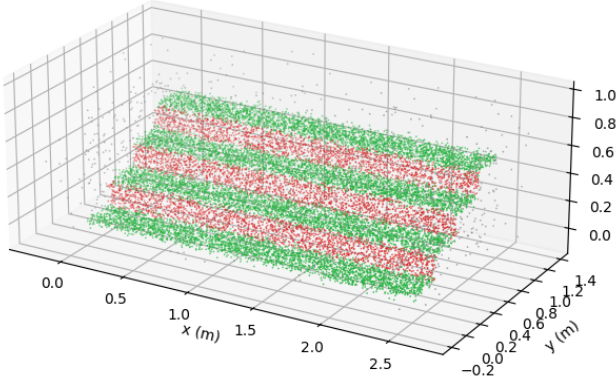


Fig. 1. Synthetic 4-step staircase point cloud, coloured by ground-truth label (green: steppable tread, red: riser, grey: other). $\sim 18k$ points with LiDAR noise and outliers.

- 3) *Edge clutter*: a thin band of points at each tread/riser join, labelled *other*, emulating range-discontinuity artefacts.

The default setup yields 18,270 points (12,000 steppable, 5,400 riser, 870 other). The generator is implemented in NumPy [8]; clouds can be exported for interactive inspection in Open3D [9]. Each point has a specific label, so that it may be evaluated quantitatively. A typical raw cloud is seen in Fig. 1.

III. METHODOLOGY

All four methods output a per-point label in $\{\text{STEPPABLE}, \text{RISER}, \text{OTHER}\}$. A surface is deemed horizontal/vertical by the angle between its (unit) normal $\mathbf{n}$ and the z -axis: with $n_z = |\mathbf{n} \cdot \hat{\mathbf{z}}|$, a surface is horizontal if $n_z \geq \cos \theta_h$ and vertical if $n_z \leq \cos \theta_v$, for thresholds θ_h, θ_v .

A. PCA Normal Analysis

For each point we gather its k nearest neighbours and form the local covariance

$$C = \frac{1}{k} \sum_{j=1}^k (\mathbf{p}_j - \bar{\mathbf{p}})(\mathbf{p}_j - \bar{\mathbf{p}})^\top, \quad (3)$$

Its eigenvector associated to the weakest eigenvalue estimates the surface normal $\mathbf{n}$ [4]. Then points are directly classified by n_z . This is the most local method and it is the most susceptible to surface jitter, but it does not need a global model. We utilize $k = 50$ neighbors, computed with the nearest-neighbor search of scikit-learn [5].

B. Multi-Plane RANSAC

We iteratively fit planes using RANSAC [6] and discard inliers after each fit. The top edge of a riser is almost coplanar in z with the tread above it, so a naïve horizontal-first fitting

TABLE I
STEPPABLE-CLASS METRICS ON THE DEFAULT CLOUD (18,270 POINTS).
BEST PER COLUMN IN BOLD.

Method	Acc	P	R	F1	IoU
PCA normals	0.794	0.966	0.865	0.913	0.840
Height-histogram	0.722	0.975	0.818	0.889	0.801
DBSCAN + slope	0.769	0.977	0.796	0.877	0.782
RANSAC multi-plane	0.711	0.888	0.701	0.784	0.644

leaks riser points into treads. We thus first extract *vertical* planes and then *horizontal* planes with a tight orientation tolerance ($\theta_h = 15^\circ$, $\theta_v = 75^\circ$) so that no planes near the oblique staircase incline ($\approx 31^\circ$) are permitted. A plane is fitted from a minimal sample and scored by its number of inliers with a distance criterion of $\tau = 0.04$ m; planes with less than a minimum number of inliers are rejected.

C. Normal-Augmented DBSCAN with Slope Analysis

Raw- xyz DBSCAN [7] collapses the whole staircase into one connected component because adjacent treads and risers touch. We instead cluster in the augmented feature space

$$\phi(\mathbf{p}) = [x, y, z, w_n n_x, w_n n_y, w_n n_z], \quad (4)$$

so two perpendicular but spatially near surfaces are separated by their different normals. We utilize smoothed normals ($k = 80$) and weight $w_n = 2.0$. We use $\varepsilon = 0.20$ and $\text{min_samples} = 20$. Each resulting cluster is labeled by the normal of its globally fitted plane. DBSCAN noise points become *other*.

D. Height-Histogram Tread Detection

Treads are located in narrow, densely populated bands in z . We histogram z with 1 cm precision, find local maxima above an adaptive prominence threshold, and mark locations within a band of half-width 0.04 m about each peak as potential treads, refined by the local normal. This is a fast solution, and makes use of the regularity of the staircase directly.

IV. EXPERIMENTAL RESULTS

We evaluate the steppable class with Accuracy, Precision, Recall, F1, and IoU. Precision and Recall are $P = \frac{TP}{TP+FP}$, $R = \frac{TP}{TP+FN}$, with $F1 = \frac{2PR}{P+R}$ and $\text{IoU} = \frac{TP}{TP+FP+FN}$, where a positive is a point predicted STEPPABLE. Table I reports all methods on the default cloud (seed fixed for reproducibility).

The PCA-normal approach has the best F1, IoU and best recall, whereas DBSCAN and the height-histogram have the highest steppable precision (≈ 0.98). The before/after segmentation of the best technique is shown in Fig. 2 and the confusion matrix is shown in Fig. 3. The primary error mode is steppable points being absorbed into *other* near tread edges, rather than confusion with risers.

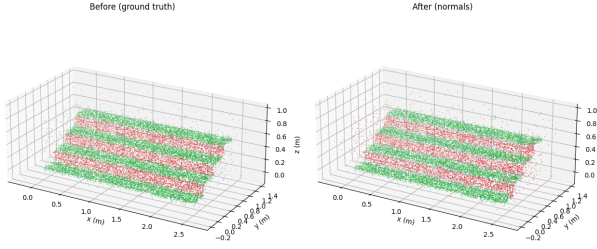


Fig. 2. Before/after segmentation by the PCA-normal method. Left: ground-truth labels; right: predicted labels (green: steppable, red: riser, grey: other).

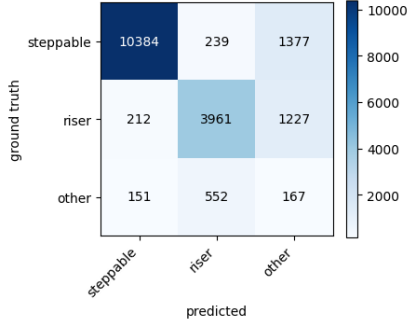


Fig. 3. Confusion matrix of the PCA-normal method; rows are ground truth, columns predictions, ordered steppable / riser / other.

A. Noise and Outlier Sensitivity

We sweep the LiDAR noise σ_L and outlier ratio ρ and record the steppable F1 (Fig. 4). F1 drops from 0.918 at $\sigma_L = 0$ to 0.861 at $\sigma_L = 0.02$ and 0.577 at $\sigma_L = 0.05$, showing local-normal quality as the limiting factor: even the mandated per-step jitter $\sigma_z = 0.02$ m substantially reduces normals. Robustness to outliers is much stronger: F1 only decreases from 0.918 to 0.875 when ρ increases from 0 to 0.20, since outliers rarely form coherent planar support.

V. ANALYSIS AND DISCUSSION

Which metric matters for safety? The cost of errors for a stair-following robot is not symmetric. Dangerous is the step when tread is wanting; not to skip a proper tread is not. Hence, the key objective is high *precision* of the steppable class, possibly at the expense of recall. The DBSCAN and height-histogram approaches are attractive operating points under this criterion at $P \approx 0.98$, even though their F1 is below the PCA-normal method: a robot employing their output would nearly never be directed to step on a non-tread.

Local vs. global geometry. The PCA-normal approach is fully local, and hence inherits the full surface jitter. This is why the PCA-normal method has good recall (it identifies virtually all genuine treads) but is sensitive to noise. The plane- and cluster-based approaches are more cautious, more precise, since they aggregate evidence over multiple sites.

Is it safe enough? A precision-driven approach results in very few hazardous steppable calls for clean to moderate noise ($\sigma_L \leq 0.02$ m), which, when paired with a planner that selects only high-confidence, spatially consistent footholds, forms a

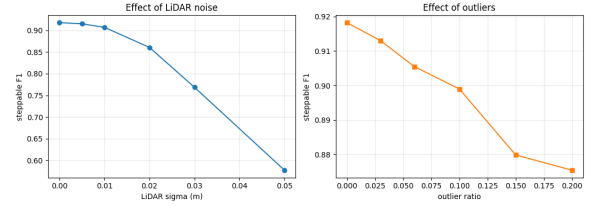


Fig. 4. Steppable F1 vs. LiDAR noise σ_L (left) and outlier ratio ρ (right) for the PCA-normal method.

reasonable basis for a cautious controller. At high noise ($\sigma_L \geq 0.05$ m) both precision and recall drop and a single geometric pass is no longer enough; temporal fusion across frames or a learnt refining stage would be required.

Limitations. The data is generated and the staircase is regular and not obstructed. Real scans introduce self occlusion, non-uniform density, specular dropouts and clutter (e.g. railings, debris). The methods described here provide a visible and reproducible baseline against which such impacts, including learning-based methods, can be tested.

VI. CONCLUSION

We provide a replicable comparison of four conventional geometric approaches for the segmentation of steppable surfaces in a synthetic LiDAR staircase point cloud. Overall, the PCA-normal technique performed the best (F1 = 0.913, IoU = 0.840), whereas normal-augmented DBSCAN and a height-histogram detector produced the highest steppable precision (≈ 0.98), which we claim is the deciding metric for foothold safety. The noise analysis indicates that the binding restriction was local-normal quality, not outliers. Future work involves evaluation on real scans, temporal fusion and a hybrid that uses a high accuracy geometric prior to supervise a learnt segmenter.

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